Inkloom AI

• FINAL HACKATHON PROJECT SUBMISSION

Intelligent Brand Governance & Publishing Platform

A complete pre-publication quality and compliance system. Inkloom AI audits marketing text, visual banners, and multi-slide pitch decks against company rules and official technical facts before publication. It flags forbidden words, verifies facts, offers 1-click intent-preserving rewrites, enables team approvals, and publishes approved content directly to social media and external CMS platforms.

1. TEXT & TONE AUDIT
Sub-50ms regex and Gemini RAG checking voice, readability, and banned words.

2. PITCH DECK SCANNER
Slide-by-slide checks for logo margins (<20px), all-caps fonts, and slide claims.

3. 1-CLICK PUBLISHING
Zero-API Twitter & LinkedIn Web Intents plus Universal Outbound Webhooks.

4. SELF-HEALING RULES
AI feedback loop clarifying frequently broken rules with Do's and Don'ts.

TEAM DETAILS
SATYACODEZ
FutureStack Hackathon

TEAM LEADER
Satyam Kumar
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VERIFIED SOURCE CODE
GitHub Public Repository
`satya-no17/futureStack`

The Complete Shipped Feature Suite

A Fully Integrated Pre-Publication Ecosystem Working Across 8 Unified Dashboard Modules

01

Pre-Publication Review

Automated multi-stage analysis checking drafts for policy compliance before any live release.

- **Banned Terminology:** Catches legal risk words like "guaranteed" or "cheap".
- **Voice Consistency:** Flags hype, passive voice, and aggressive slang.
- **Truth Verification:** Verifies stats against technical fact sheets.
- **Readability:** Scores Flesch-Kincaid grade level and sentence length.

Sub-50ms
LOCAL RULE EVALUATION LATENCY

02

Intent-Preserving Fixes

Pinpoint suggestions that improve brand alignment without wiping out the writer's authentic tone.

- **Token Highlighting:** Color-codes exact character ranges in draft text.
- **Non-Destructive:** Fixes single phrases rather than rewriting whole paragraphs.
- **1-Click Replace:** Swaps in approved vocabulary instantly.
- **Live Score Update:** Recalculates 0–100 score immediately upon fix.

100%
OFFSET-GROUNDED TEXT FIXES

03

Team Review Queue

Human-in-the-loop governance interface coordinating writers, editors, and brand leads.

- **Approval Stages:** Draft → In Review → Approved / Changes Requested.
- **Side-by-Side Diffs:** Shows exact red deletions and green additions.
- **Threaded Comments:** Contextual annotations attached to text lines.
- **Role Permissions:** Creator, Reviewer, and Admin capabilities.

3 Roles
ROLE-BASED ACCESS CONTROL (RBAC)

04

Round 3 Commitments

Delivered all three promised major capabilities fully verified in the live application.

- **Slide Deck Scanner:** Multi-slide layout margins, logo safety & claims.
- **Social Publishing:** Zero-API Twitter/X & LinkedIn intent dispatch.
- **Universal Webhooks:** Signed payloads to WordPress, Webflow, Zapier.
- **Self-Healing Rules:** AI auto-clarification of repeated violations.

3 of 3
MAJOR COMMITMENTS COMPLETED

✓ **Live Dashboard Modules: Draft Studio • Review Queue • Slide Deck Scanner • Guideline Vector Store • Technical Fact Sheets • Analytics Dashboard • Self-Healing Rules • Settings**

ALL 8 TABS OPERATIONAL

The 4-Tier Hybrid Content Review Pipeline

Multi-Dimensional Analysis Delivering Explainable, Fact-Grounded Findings in Milliseconds

Tier 1 • Instant Filter

Banned Terminology

High-speed regex engine matching forbidden vocabulary, liability triggers, and outdated marketing claims.

- **Prohibited Terms:** "cheap", "guaranteed", "#1 best", "instant", "faultless".
- **Word Boundaries:** Regex word boundary logic prevents false positive matches.
- **Severity Weighting:** Flags categorized as High, Medium, or Low impact.
- **Recommended Terms:** Pairs every banned word with approved brand phrasing.

< 5ms
EXECUTION SPEED

Tier 2 • Semantic RAG

Voice & Tone Grounding

Retrieves top guideline vector chunks via cosine similarity to analyze nuance, empathy, and professional style.

- **Tone Analysis:** Catches aggressive sales pressure, hyperbole, and slang.
- **Guideline Citation:** Explains exactly which brand rule was breached and why.
- **Perspective Checks:** Enforces first-person plural ("We build") vs impersonal voice.
- **Vector Retrieval:** Queries 384-dimensional dense embeddings in pgvector.

Top-4
VECTOR CHUNKS MATCHED

Tier 3 • Fact Verification

Claim Truth Checking

Cross-examines promotional statistics, pricing, and technical claims against indexed company fact sheets.

- **Tri-State Scoring:** Supported (verified), Unverified (no proof), Contradicted (false).
- **SLA & Compliance:** Confirms uptime claims (e.g. 99.95%) and SOC2 statements.
- **Source Attribution:** Names the approved reference document for every check.
- **Anti-Hallucination:** Strict prompt constraints prevent false AI assumptions.

Tri-State
VERIFICATION CLASSIFICATION

Tier 4 • Readability & Visual

Accessibility & Scoring

Computes readability grade levels and verifies image contrast standards for visual accessibility.

- **Flesch-Kincaid:** Computes reading grade level (optimal target: Grade 8 to 11).
- **Sentence Length:** Flags dense sentences exceeding 30 words for clarity.
- **WCAG 2.1 AA/AAA:** Audits visual banner color contrast ratios (min 4.5:1).
- **Weighted Score:** Generates unified 0 to 100 overall compliance score.

WCAG AA
CONTRAST RATIO STANDARD

Draft Studio: In-Line Token Highlighter & 1-Click Fixes

Precision Suggestions That Preserve the Author's Voice While Correcting Off-Brand Words

In-Line Visual Highlighter Mechanics

Draft text renders with non-overlapping color-coded visual tokens. Each token marks the exact character offsets of an identified issue:

```
// Live draft representation in Inkloom Draft Studio: "Our crazy fast cloud is guaranteed to cut costs by 80% with instant setup." • Red Token: Banned Term ("crazy fast" → suggests "sub-50ms latency") • Yellow Token: Unverified Claim ("guaranteed" → suggests "backed by 99.95% SLA") • Orange Token: Tone Deviation ("instant" → suggests "near real-time")
```

- **Exact Character Offsets:** Bounded by `startIndex` and `endIndex` to prevent text jumping.
- **Direct Inline Editing:** Writers can edit text directly in place, then click "Check My Words" to re-evaluate in real time.
- **Selective Correction:** Click any highlighted phrase to inspect the specific rule and reason.

Intent-Preserving Correction Cards

Generic AI tools rewrite entire paragraphs destructively. Inkloom isolates only the offending token and offers a 1-click replacement:

MUST FIX • BANNED TERM

Source: Brand Policy #14

Problem phrase: “cheap cloud storage”

The word "cheap" conveys low reliability and compromises enterprise brand positioning.

Fix: “cost-effective cloud storage” 1-Click Apply

- **Dynamic Score Update:** The document score increments immediately upon accepting a suggested fix.
- **Compliance Dismissal Note:** Writers who reject a suggestion must submit a justification note, preserving audit accountability.
- **Tone Preservation:** Sentence cadence and narrative voice remain 100% authored by the creator.

Editorial Review Queue & Side-by-Side Diff Viewer

Structured Human-in-the-Loop Governance for Distributed Marketing Teams and Agencies

Editorial Review Queue

Submitted drafts transition into the Review Queue, providing marketing leads and reviewers with clear action stages:

- **Stage Progression:** Moves content cleanly from `Draft` → `In Review` → `Approved` or `Changes Requested`.
- **Reviewer Decision Controls:** Reviewers can approve compliant drafts or request revisions with structured feedback.
- **Threaded Annotations:** Leave line-item feedback directly tied to specific paragraphs or claims.
- **Audit Trail Logging:** Immutable timestamped records of who submitted, revised, and approved every asset.

100%
LOGGED AUDIT HISTORY

Instant
HANDOFF TO REVIEWER

Side-by-Side Version Diff Viewer

Reviewers can inspect exact word-by-word differences between draft revisions, saving hours of manual proofreading:

REMOVED IN REVISION (RED):

```
- "Our crazy fast platform is the cheapest on the market guaranteed."
```

APPROVED REVISION (GREEN):

```
+ "Our platform delivers sub-50ms latency backed by our 99.95% uptime SLA."
```

- **Version History:** Step back through any historical snapshot with full restore capability.
- **Resolution Tracking:** Highlights whether requested changes were addressed or dismissed.
- **Publish Gate:** Prevents live publication until an authorized reviewer signs off.

ROUND 3 COMMITMENT 1 DELIVERED

Multi-Modal Slide Deck Layout & Claim Scanner

Delivered: Automated Pitch Deck Ingestion, Layout Validation & Factual Spec Auditing

1. Presentation Ingestion & Parsing

Parses structured multi-slide decks and PPTX XML layout trees, breaking decks down into individual slide nodes:

- **Slide Segmentation:** Extracts slide numbers, titles, body paragraphs, and visual bounding boxes.
- **Coordinate Mapping:** Captures `(x, y, width, height)` coordinates for text boxes and image elements.
- **Slide-by-Slide Scoring:** Computes an independent compliance score (0–100) for every slide.
- **Thumbnail Carousel:** Visual slide selector in the UI for rapid navigation across the deck.

2. Layout Margins & Safety Zones

Audits visual boundaries, brand logo exclusion margins, and visual typography guidelines:

- **Logo Clearance Perimeter:** Automatically flags any element violating the <20px edge safety clearance.
- **Typography Violations:** Detects aggressive ALL-CAPS headlines (>4 consecutive capital words).
- **Element Crowding:** Flags overlapping text boxes and poorly positioned graphics.
- **Visual Hierarchy:** Checks title-to-body font scale ratios across all slides.

3. Slide Claim RAG Verification

Cross-examines bullet-point claims and metrics on slides against approved technical fact sheets:

- **Superlative Claims:** Flags unsubstantiated marketing phrases like "#1 fastest" or "instant setup".
- **Metric Grounding:** Verifies uptime numbers (99.95%) and SLA guarantees against spec sheets.
- **Slide Action Cards:** Generates targeted correction cards directly attached to each offending slide.
- **Verified Pass Status:** Decks pass only when all slides meet minimum compliance thresholds.

✓ **Verified in Automated Test Suite: Tested a 3-slide pitch deck with typography violations, boundary crowding, and unverified claims (caught 13 issues, score: 54/100).**

VERIFIED WORKING

ROUND 3 COMMITMENT 2 DELIVERED

Zero-API Social Publishing & Webhook Connectors

Delivered: 1-Click Publishing to Social Networks and Outbound Headless Webhooks

Zero-API Social Web Intent Publishing

Judges and users can publish approved content immediately without needing developer API keys, OAuth credentials, or third-party approvals:

- **Twitter / X Web Intent:** Pre-formats approved text, validates character count against the 280-char limit, and launches a 1-click posting window (`twitter.com/intent/tweet`).
- **LinkedIn Share Intent:** Automatically formats title, summary, and hashtags into LinkedIn's share interface (`linkedin.com/sharing/share-offsite`).
- **Compliance Trust Badge:** Appends optional verifiable watermark: `[Verified by Inkloom AI — Score 100/100]`.
- **7 Target Channels:** Twitter/X, LinkedIn, WordPress, Webflow, Shopify, Ghost, and Custom Webhooks.

7 Channels
SUPPORTED DESTINATIONS

Zero Setup
NO API KEYS REQUIRED

Universal Headless Webhook Dispatcher

Pushes approved, compliant drafts directly to external CMS platforms (WordPress, Webflow) or automation pipelines (Zapier, Make, Slack):

```
// Outbound Webhook JSON Contract: { "event": "content.published", "contentId": "c4b18e-49d4-9f15-published", "title": "Enterprise Cloud Infrastructure Launch", "score": 100, "status": "approved", "publishedBy": "reviewer@company.com", "timestamp": "2026-10-30T22:15:00Z", "verificationSignature": "sha256-hmac-inkloom-valid" }
```

- **Custom Authorization:** Supports custom Bearer tokens and API key headers.
- **Safe Execution:** 10-second timeout safeguard with automated delivery verification logging.
- **Audit Preservation:** Persists webhook response codes and delivery status in PostgreSQL.

ROUND 3 COMMITMENT 3 DELIVERED

Self-Healing Brand Rules & Ambiguity Clarifier

Delivered: Continuous AI Clustering That Auto-Refines Confusing Brand Guidelines

1. Anomaly & Ambiguity Clustering

Continuously analyzes audit logs and review history to identify brand rules that team members frequently fail:

- **Failure Tracking:** Counts repeated violation frequencies per guideline rule chunk.
- **Violator Diversity:** Identifies rules failed by multiple independent team members.
- **Dismissal Detection:** Flags rules where writers frequently dismiss AI suggestions with notes.
- **Ambiguity Radar:** Automatically surfaces problematic rules on the Dashboard.

2. AI-Driven Rule Clarification

Queries Google Gemini with the ambiguous guideline chunk and real failure cases to draft clear instructions:

- **Actionable Phrasing:** Replaces vague advice ("Be punchy") with explicit, actionable boundaries.
- **Do & Don't Pairs:** Generates concrete pairs of acceptable and prohibited writing examples.
- **Root Cause Analysis:** Explains exactly why creators struggled with the original wording.
- **Review Ready:** Delivers clean, ready-to-approve revision cards to marketing leads.

3. 1-Click Rulebook & Vector Update

Marketing leads can review the suggested clarification and apply it to the live governance engine in one click:

- **Instant DB Sync:** Appends or updates the clarified rule in the PostgreSQL database.
- **Vector Re-indexing:** Re-chunks text and updates vector embeddings in pgvector.
- **Zero Restart:** All future content scans immediately enforce the newly clarified rule.
- **Revision History:** Maintains an immutable log of who approved guideline changes.

✓ **Live on the Score & Reports Tab: SelfHealingCard displays active AI recommendations with a 1-click "Apply Clarification" button that updates the rule store.**

VERIFIED WORKING

ENGINEERING DEPTH

Technical Architecture & Infrastructure

Modular TypeScript Microservice, Vector-Indexed Cloud Storage, and Resilient AI Failover

Frontend Layer

Next.js 16 + React 19 + Tailwind CSS

App Router architecture with Turbopack compilation. Features high-contrast light theme, React Error Boundaries, and offline fallback interceptors.

• Next.js 16 App Router & React 19 • Tailwind CSS custom design tokens • Character-level text highlight engine • Interactive multi-slide preview modal • Lucide React icon suite

Backend API Layer

Node.js + Express + TypeScript

High-throughput REST API running on port 4000. Coordinates deterministic regex rule checks and vector RAG retrieval with SHA-256 payload caching.

• Modular Express route controllers • Zod schema validation on all inputs • PPTX XML slide parsing engine • JWT auth & bcrypt password hashing • Sub-50ms deterministic regex scanner

Database & Vector Store

PostgreSQL + pgvector Extension

Relational storage managing drafts, reviews, comments, and audit logs. Executes sub-10ms cosine similarity searches across brand guideline embeddings.

• 384-dimensional dense vectors • Cosine distance search (⟺ operator) • Drizzle ORM schema migrations • Multi-role permissions schema • Audit log persistence tables

Google Gemini 1.5 Flash AI with Deterministic Heuristic Failover

Powered by `Google Gemini 1.5 Flash` for semantic voice analysis, claim grounding, and self-healing rule synthesis. If network outages occur or API keys are missing, the system gracefully falls back to deterministic rule checks (banned words, grade level, layout margins) with a visible "Rule Checks Only" badge. Zero white-screen errors.

Testing Verification & Future Product Roadmap

100% Pass Rate Across 42 Automated Scenarios and Long-Term Development Vision

Comprehensive Testing Verification

Verified through an automated 4-tier resilience suite (`verify-resilience.mjs`) and Round 3 module tests:

- **Tier 1: Feature Coverage (24/24 PASS):** Verified error boundaries, offline fallback data across 8 tabs, walkthrough steps, tooltips, and WCAG AA contrast.
- **Tier 2: Boundary & Stress (8/8 PASS):** Network timeouts, HTTP 500 recovery, session corruption, and extreme 50,000-character inputs.
- **Tier 3: Offline Reliability (6/6 PASS):** Full dashboard navigation and 1-click suggestion replacements without database or internet.
- **Tier 4: E2E User Journeys (4/4 PASS):** Creator Drafting, Reviewer Sign-Off, Outage Recovery, and Keyboard Navigation.
- **Round 3 Features (4/4 PASS):** Verified Slide Deck Scanner, Zero-API Publishing, Universal Webhook dispatch, and Self-Healing rules.

42 / 42 PASS (100%)
AUTOMATED VERIFICATION SCENARIOS PASSING

Where It Goes Next (Product Roadmap)

Strategic product expansion plan for post-hackathon commercial deployment:

- **Phase 1 (Q1) — In-Context Browser & Figma Extension:** Chrome/Edge extension bringing Inkloom's rule engine directly into Google Docs, Notion, Canva, and Figma canvas without tab switching.
- **Phase 2 (Q2) — Two-Way CMS Synchronization:** Bi-directional webhooks with WordPress, Webflow, and Contentful to automatically flag live websites when company facts or pricing change.
- **Phase 3 (Q3) — Multi-Language Brand Voice:** Grounding brand tone across global marketing teams (Spanish, German, French, Japanese) using multilingual dense vector embeddings.
- **Phase 4 (Q4) — Enterprise SSO & SOC2 Compliance:** SAML/Okta single sign-on, immutable SOC2 audit log exports, and multi-tenant workspace isolation.

4 Phases
STRUCTURED COMMERCIAL ROADMAP